CivicLens: A Citizen-Centric Platform for Constituency Accountability and Civic Transparency

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***Abstract*—Citizens can usually register a civic complaint, but the later stages of the service journey such as identifying the responsible authority, understanding progress, confirming that a claimed repair is real, and relating recurring problems to public works. This paper presents CivicLens, a citizen-centric platform that integrates issue reporting with evidence and location; category-and-jurisdiction routing to a visible authority; a timestamped status timeline; citizen verification of claimed resolutions with reopening and escalation; recurring-issue hotspot analysis; and a constituency-level oversight view. A source-labelled public-project module is specified for linking issues to project and expenditure records. The design separates departmental execution responsibility from elected-representative oversight and treats AI assistance as optional. A working prototype using OpenStreetMap-based mapping and 30 prototype complaint records for part of Chennai was inspected through screenshots of its landing page, civic map, complaint summary and representative dashboard. The evaluation reported here is limited to this walkthrough: no routing-accuracy test, user study or field deployment has yet been performed, and a protocol for those measurements is specified. The contribution is the integration of these functions in one auditable workflow together with an honest evidence-status record, not the novelty of any single feature.**

***Keywords*—civic technology, grievance redressal, citizen participation, public accountability, geospatial analytics, transparency, complaint tracking, citizen-verified resolution**

I. Introduction

Potholes, uncollected garbage, drainage failures, water leaks and broken streetlights are routine civic problems. For the citizen who reports one, the service journey has several parts: submitting the complaint, learning which authority is responsible, understanding what has happened since, knowing whether the problem was actually fixed, disputing a closure that does not match reality, and understanding how a recurring problem relates to public development spending. Registering a complaint addresses only the first part of this journey; it does not by itself provide end-to-end accountability.

Digital citizen-participation applications are an active research area, and a systematic review of 76 studies found that reporting problems to city authorities is a leading purpose of such applications, while only a minority were evaluated by end users in real-world conditions and few described data-quality or privacy mechanisms [1]. Operational systems already provide parts of the journey. India's CPGRAMS allows feedback and an appeal after a grievance is disposed [2]; the Swachhata-MoHUA app supports photo-and-location sanitation complaints, status updates with a resolution image, and reopening [3]; and FixMyStreet sends map-located reports to the responsible authority and publishes them [4]. The question addressed here is therefore not whether individual features exist, but whether routing visibility, auditable status history, citizen verification with escalation, recurring-issue analysis and source-labelled public-project context can be combined in one citizen-facing workflow and evaluated honestly.

CivicLens also encodes a deliberate division of responsibility. The responsible department remains accountable for executing a repair; an elected representative's office is given an oversight and escalation view rather than being presented as the operational authority for every complaint.

A. Problem Statement

How can a citizen-facing platform make the journey from a reported civic issue to a verified outcome visible and auditable—showing the responsible authority, recording every action, letting the citizen confirm or dispute a claimed resolution, exposing recurring problems, and relating them to source-labelled public-project information—without exposing private data or conflating departmental execution with representative oversight?

B. Objectives and Contributions

The objectives of the work are: (O1) give every complaint a unique identifier, a visible responsible authority and a timestamped history; (O2) require a citizen verification step, with reopening and escalation, before a claimed resolution is treated as final; (O3) derive recurring-issue and constituency-level views from the underlying complaint records; (O4) present project and expenditure information only with source and verification labels; and (O5) enforce role-based access so that public pages do not expose private information.

The contributions are: (1) an integrated workflow and role model connecting reporting, routing, work, verification, escalation and public analytics; (2) a lifecycle and verification design in which status changes and verification outcomes are recorded as events; (3) a prototype whose civic map, complaint summary, hotspot clustering and representative dashboard are demonstrated by screenshots; and (4) an evidence-status record that separates demonstrated, specified and optional functions, together with an evaluation protocol. Section II reviews related work, Section III describes the architecture, Section IV the design and implementation status, Section V the evaluation, and Sections VI–VIII limitations, future work and conclusions.

II. Related Work

A. Civic Complaint and Grievance Systems

CPGRAMS is the central grievance portal of the Government of India; after a grievance is disposed, the citizen may submit feedback and file an appeal, and the disposal is treated as closed only once any appeal has been decided [2]. Swachhata-MoHUA forwards a photographed, geo-located sanitation complaint to the city corporation, assigns it to a ward-level inspector, notifies the citizen with a "resolved" image, and lets the citizen reopen the complaint [3]. FixMyStreet takes a different approach: the reporter marks a location on a map and the service sends the report to the responsible council, with categories set by each council and reports displayed publicly [4]. Where councils integrate through the Open311 interface, work updates can flow back to the public site [5]. Open311 GeoReport v2 itself standardizes service-request exchange and models request status as either open or closed [6].

Taken together, these sources show that routing, notifications and post-resolution feedback or reopening already exist in separate systems. They do not, on the evidence consulted, describe a verification step as an explicit lifecycle state with three outcomes, nor recurring-issue analysis or project-expenditure context tied to the complaint record; these capabilities are marked "not established from source" in Table II rather than assumed absent.

B. Civic Transparency and Open Data

The Open Contracting Data Standard structures public-contract information by stage—planning, tender, award, contract and implementation—and uses a process identifier to join records across stages [7]. It informs the structure of the project and financial records proposed for CivicLens but is not a complaint system. A review of experimental evaluations of transparency-and-accountability programmes found mixed results and attributed variation to obstacles such as collective action, political resistance and long implementation chains [8]. This motivates pairing published information with an explicit response path—reopening and escalation—rather than relying on disclosure alone.

C. Geospatial Issue Mapping and Analytics

Map-based reporting is established practice in FixMyStreet [4]. OpenStreetMap provides freely usable, community-edited map data [9], which makes it suitable for a prototype that must avoid paid mapping services. The smart-city review notes that data quality and privacy handling are infrequently reported in citizen-participation applications [1], which is why CivicLens treats privacy-safe public views as a design requirement (Section III-E). The recurring-issue view in CivicLens groups complaints that occur close together; its parameters are described in Section IV-F, and its effectiveness has not been benchmarked.

D. Citizen Feedback and Resolution Verification

Feedback and appeal after disposal [2] and reopening after resolution [3] show that closing a case without checking the outcome is recognised as a problem. A synthesis of social-accountability evidence reports mixed impacts, finds bounded information-only interventions weaker than strategies that also strengthen the state's capacity to respond, and describes the need for citizen "voice" to be paired with institutional "teeth" [10]. In CivicLens this is reflected by recording the citizen's verdict as an event and routing disputed or overdue cases to a higher authority; whether this improves outcomes is an empirical question not yet tested.

E. Research Gap

CivicLens investigates the integration of citizen reporting, visible authority routing, an auditable status history, citizen verification with reopening and escalation, recurring-issue analysis, and source-labelled public-project context in one citizen-facing workflow. We do not claim that any single feature is new, nor that no comparable integrated system exists; the sources reviewed document these capabilities in separate systems, and a broader systematic review is needed before a stronger novelty claim could be made. The contribution is positioned as integration and evaluation (Table II).

TABLE II
COMPARISON OF RELATED SYSTEMS AND STUDIES

System / study	Core capability	Status / feedback visibility	Geospatial / recurring-issue support	Project transparency	Limitation / relevance
CPGRAMS [2]	Central online grievance registration and monitoring across public authorities	Feedback and appeal after disposal; call-centre feedback	Not established from source	Not established from source	Local street-level mapping not established; relevant for post-closure feedback
Swachhata-MoHUA [3]	Photo and location complaints routed to the city corporation, assigned to a ward inspector	Push updates with "resolved" image; citizen may reopen	Location captured with photo; recurrence analysis not established	Not established from source	Listed categories are sanitation-focused

System / study	Core capability	Status / feedback visibility	Geospatial / recurring-issue support	Project transparency	Limitation / relevance
FixMyStreet [4], [5]	Map-pin reports sent to the responsible council; categories set by councils	Reports shown publicly; updates returned via Open311 where integrated	Map-based; recurrence analysis not established	Not established from source	UK authority structure; relevant for routing and public visibility
Open311 GeoReport v2 [6]	Interoperable API for service requests and service lists	Request status modelled as open or closed	Request coordinates in the data model	Not applicable	Two-valued status; verification semantics not specified in the cited material
OCDS [7]	Open data standard for contracting stages, planning to implementation	Not applicable	Not established from source	Stage-wise public-contract data joined by a process identifier	Not a complaint system; informs project-record design
Smart-city participation review [1]	Systematic review of 76 studies of citizen-participation applications	Few studies evaluated by end users in real-world use	Covers reporting, sensing and analysis approaches	A minority address decision-making and administrative transparency	Review, not a system; motivates evaluation and privacy focus
CivicLens (this work)	Integrated report → route → assign → work → verify → escalate → analyse workflow	Three-outcome citizen verification; reopening; timeline (specified, partly observed)	OpenStreetMap-based map; hotspot clustering (observed)	Source-labelled records (specified; not evidenced)	Prototype; no user study or field validation yet

III. CivicLens System Architecture

A. Overall Architecture

CivicLens is organised in four layers (Fig. 1). A responsive web interface presents role-specific views to citizens, officers, field workers, representatives, administrators and public viewers. An authentication and role-based access layer controls what each role may see and do. The service layer comprises issue reporting, location and authority routing, workflow and timeline, citizen verification, escalation and notifications, public map and hotspot analytics, the constituency dashboard, project and fund transparency, an audit log and an optional AI assistant. A data layer holds issue, assignment and timeline records, evidence files, a jurisdiction and department registry, and project and financial records. OpenStreetMap tiles are the external map source; official project data feeds are not assumed to be available.

Fig. 1 distinguishes components observed in the prototype screenshots, components that are specified but not evidenced in the material supplied, and optional components. A conceptual diagram prepared during design review proposed a cloud-hosted microservice deployment, whereas the project specification lists a free-only stack; neither is confirmed by code or configuration here, so Fig. 1 shows logical components only.

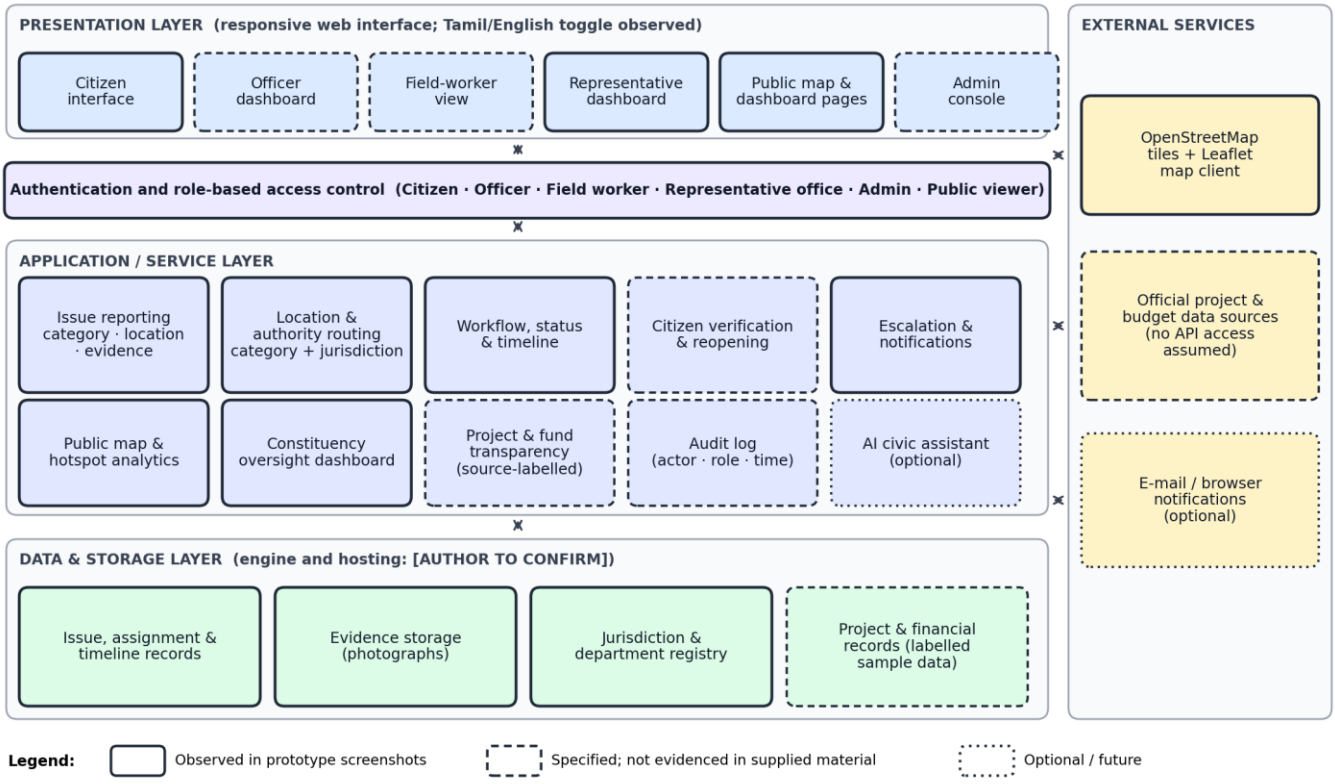


Fig. 1. Overall CivicLens architecture. Solid borders: observed in prototype screenshots; dashed: specified but not evidenced in the supplied material; dotted: optional or future. Deployment technologies are not shown because they are unconfirmed.

B. User Roles and Access Control

Six roles are defined. *Citizens* report issues, track their own complaints, supply evidence and verify resolutions. *Department officers* review, assign, update and resolve complaints within their department. *Field workers* see only the information needed for their assigned tasks and submit work updates. *Representative offices* view constituency-level information and monitor unresolved and recurring issues without acting as the department's executor. *Administrators* manage users, categories, jurisdictions, authorities, project data and rules. *Public viewers* access maps and dashboards without private records. Design-review personas additionally include low-digital-literacy citizens and persons with disabilities, motivating a local-language interface and simple flows. Role-specific views were observed: the citizen session shows "My Complaints" and a report button, whereas the representative session shows neither and opens an oversight dashboard (Fig. 4, Fig. 5).

C. Core Data Model

Table III lists the principal entities and key fields as given in the project specification; the implemented schema has not been inspected and is to be confirmed by the authors. An Issue carries a unique identifier (the format CL-2026-000130 was observed), category, description, location, severity or priority, status and timestamps. Verification outcomes, escalations and timeline events are separate records so that the history is appended to rather than overwritten.

TABLE III
MAIN DATA ENTITIES AND KEY FIELDS

Entity	Key fields (specification)	Relationships
User / Profile	id, role, preferred language, verification state	Reports Issues; role sets permissions
Issue / Complaint	id, category, description, location, severity/priority, status, timestamps	Has Evidence, Assignments, Timeline Events; lies in a Constituency
Issue Evidence	issue id, media type, storage reference, metadata	Belongs to Issue
Authority / Department	department, jurisdiction, contact or role information	Receives Assignments
Assignment	issue id, authority id, assigned and accepted times, field worker	Links Issue to Authority
Timeline Event	issue id, actor and role, action, timestamp, notes	Append-only history of Issue
Citizen Verification	issue id, result, comment/evidence, timestamp, review state	One or more per Issue
Escalation	issue id, reason, target level, timestamps	Preserves Issue history
Constituency / Jurisdiction	boundary, wards, elected offices, administrative metadata	Contains Issues and Projects
Project	name, location, agency, scheme, status, dates, source	Optionally linked to Issues
Financial Record	project id, allocation, expenditure, source, update date	Belongs to Project
Notification	recipient, event, read state, timestamp	Generated from events
Audit Log	actor, action, target, timestamp	Covers important actions

D. End-to-End Issue Lifecycle

The specified statuses are Submitted, Under Verification, Assigned, Acknowledged, In Progress, Awaiting Information, Resolved, Reopened, Escalated and Closed. Each status change should record the responsible role, a timestamp and the action taken. Fig. 2 shows the lifecycle: a report is validated and routed; if routing is uncertain, an authorised user reviews and reassigns it with an audit entry; the officer and field worker acknowledge and carry out the work and upload evidence; the officer marks the issue resolved; the citizen verifies (Section IV-E); and the case is closed, or reopened and, if disputed or overdue, escalated. Closed records feed the public analytics. The map legend of the prototype groups statuses into four classes: submitted or assigned, in progress or reopened, critical or escalated, and resolved or closed.

Priority is intended to reflect impact rather than attention: safety risk, number of people affected, vulnerable groups nearby, duration, recurrence and essential-service impact. *Illustrative example (not a documented case):* a missing manhole cover near a primary school is reported with a photograph and location; category and ward route it to the sewerage or local-body authority; an officer assigns a field worker; the worker uploads a photograph of the replaced cover; the officer marks the issue resolved; the reporter reports it "partially resolved" because the cover is loose; the case is reopened and, if the overdue timeline is exceeded, escalated with its history intact.

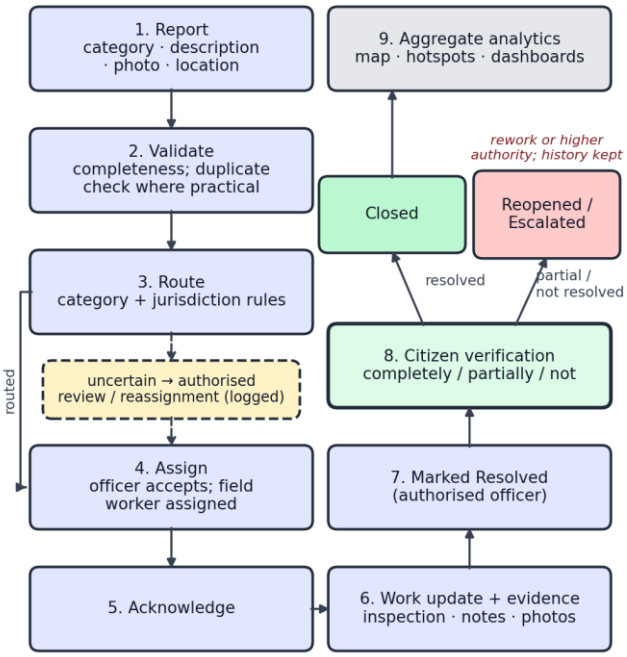


Fig. 2. End-to-end issue lifecycle. The dashed amber step is the audited fallback when jurisdiction is uncertain. Reopened or escalated cases return to work or to a higher authority with history preserved.

E. Privacy, Security, and Trust

Public pages are designed to omit phone numbers, private addresses, credentials and sensitive images; the complaint summary observed in the prototype shows ward, department, date, severity and an evidence photograph, and the reporter's identity is not displayed there (whether every public view behaves identically is to be tested). Role-based permissions apply least privilege; important actions are written to an audit

log; and verification outcomes are never silently overwritten. Anti-abuse measures specified for later work include rate limiting, duplicate detection and location validation. Evidence retention, false or abusive reports and location error remain open risks. Public financial information carries a source, an update date and a verification state, and sample data is labelled as such. AI, where used, may recommend but does not make administrative decisions.

IV. System Design and Implementation

Table I lists the twelve core modules with an implementation-status code: **A**, demonstrable in the supplied screenshots; **U**, not evidenced in the supplied material (specified, or only a navigation entry was observed) and to be confirmed by the authors; and **D**, optional future enhancement. No module is claimed as tested.

TABLE I
CORE MODULES, FUNCTIONS AND STATUS

Module	Function	Status
1 Authentication and roles	Role-based access and dashboards	A (role views); U (sign-up, reset)
2 Civic issue reporting	Category, location, evidence, severity; unique ID	A (records); U (form, voice)
3 Location and authority routing	Category + jurisdiction rules; visible ward and department; reviewed reassignment	A (output); accuracy not measured
4 Complaint tracking and timeline	Status lifecycle; timestamped events	A (status); U (timeline detail)
5 Citizen resolution verification	Resolved / partially / not resolved; reopen; escalate	U (aggregate reopen count observed)
6 Officer dashboard	Work queue, assignment, status update	U
7 Field-worker workflow	Assigned tasks; progress evidence	U
8 Public map and analytics	Filterable map; hotspot clusters	A
9 Constituency dashboard	KPIs, category and status charts	A (representative view)
10 Project and fund transparency	Source-labelled projects and expenditure	U (navigation entry only)
11 AI civic assistant	Guidance with rule-based fallback	U (navigation entry only)
12 Notifications and escalation	In-app alerts; overdue and dispute escalation	A (indicators); U (rules)

A. Civic Issue Reporting

Purpose and inputs: citizens report a problem with category, description, photographic evidence, location and severity or impact; optional voice input is specified. *Processing:* the system checks completeness and, where practical, possible duplicates, then generates a unique complaint ID and shows the initial status and responsible authority. *Observed:* records carrying a complaint ID, ward, severity and an evidence photograph, report entry points on the landing page and map, and the category set roads, garbage, drainage, water, streetlights, public facilities and other (Fig. 4, Fig. 5). The entry form itself was not among the supplied screenshots.

B. Location and Authority Routing

Routing combines the issue category with location or jurisdiction rules held in a registry, and displays the resulting department, ward and routing status to the citizen. The observed complaint was routed to the Roads and Bridges Department within Ward 176 – Guindy (Fig. 4c). Jurisdiction must be configured from authoritative administrative information for the pilot geography, and the system must not assume that a representative is the executing authority. When routing is uncertain, an officer or administrator reviews and reassigns the complaint and the action is logged. Only a severity label was observed; the impact-based priority model is a design specification.

C. Complaint Tracking and Audit Timeline

Every complaint is intended to show its current status, latest action, next expected step and, where a rule exists, an expected resolution time. Timeline events cover submission, assignment, acknowledgement, field update, evidence upload, resolution and citizen verification. The prototype shows a status label and a “View Full Timeline and Evidence” control (Fig. 4c); the content of the timeline view was not supplied.

D. Officer and Field-Worker Workflow

Officers are to see new, assigned, in-progress, overdue, reopened and resolved-awaiting-verification items, with filters by category, area, severity, status, date and worker, and actions to accept, request clarification, reassign where permitted, assign a field worker, update status, add evidence and mark resolved. Field workers follow a mobile-first sequence—view assignment, open location, inspect, update status, add evidence, submit completion—and see only what is needed. Completion returns to the officer and then to citizen verification rather than bypassing it. These modules are specified; no screenshots were supplied.

E. Citizen Verification and Escalation

When an authority marks a complaint resolved, the citizen receives an in-app request and may select completely resolved, partially resolved or not resolved, with an optional comment and new photograph (Fig. 3). A complete confirmation allows closure under the configured rule; the other outcomes return the case for reverification or rework, reopen it, or escalate it if it is disputed or overdue. The verdict is stored as an event. A single anonymous verdict is not meant to determine official resolution on its own; disputed cases receive administrative review, and escalation preserves the history and records the reason. The dashboards show one reopened and one escalated complaint, the latter described as pending commissioner action (Fig. 5), but the verification interface itself was not supplied.

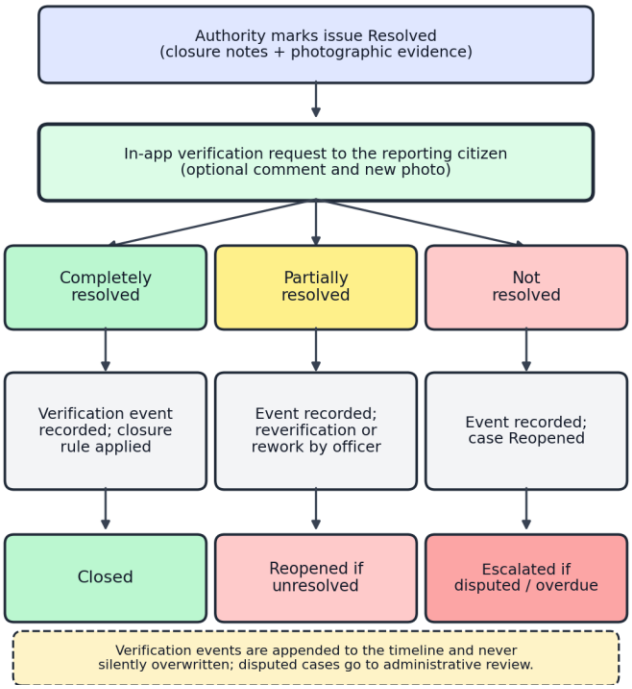


Fig. 3. Citizen verification flow. Each outcome is recorded as an appended timeline event; reopening and escalation preserve the case history.

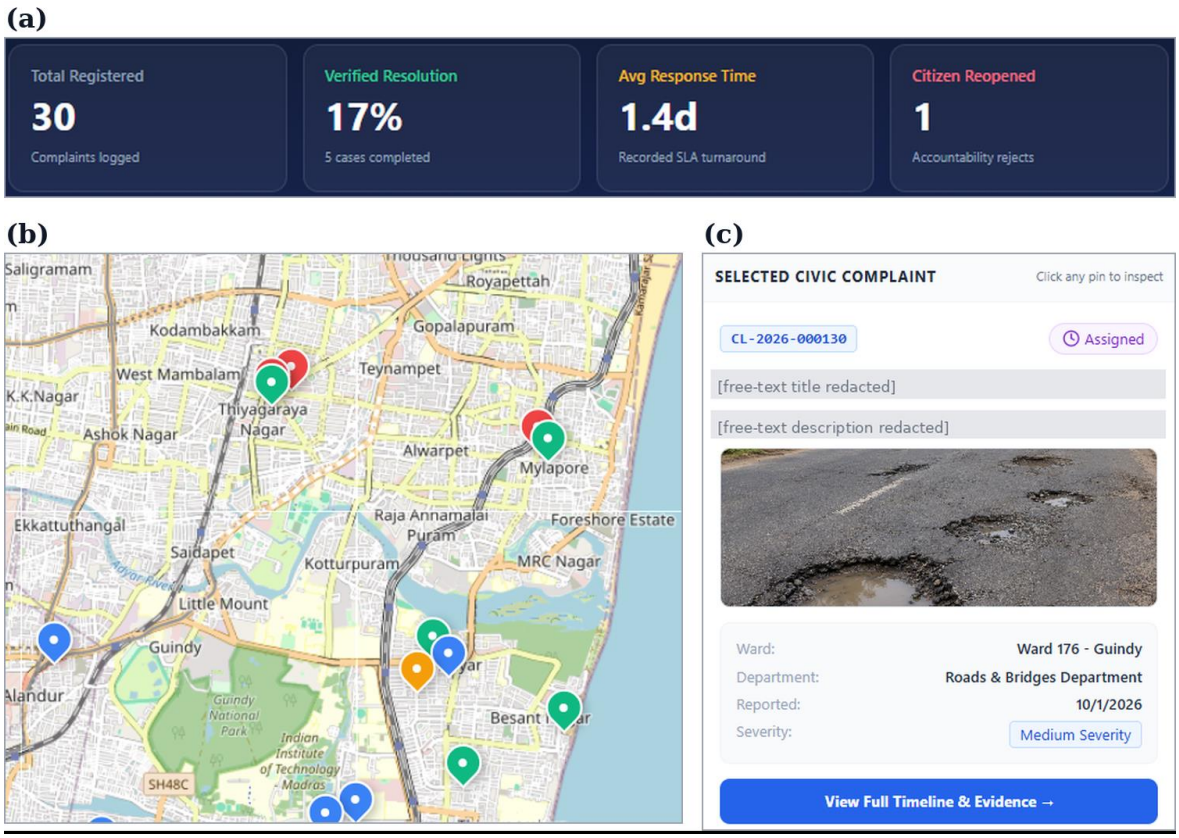


Fig. 4. Prototype screens (30 prototype complaint records, Chennai South area). (a) Landing-page counters. (b) Civic map built on OpenStreetMap with Leaflet; pin colours: green resolved or closed, amber in progress or reopened, red critical or escalated, blue submitted or assigned. (c) Selected-complaint panel showing identifier, status, ward, department, severity and evidence photograph; the free-text title and description are redacted, and the navigation bar with account names was cropped out.

F. Public Civic Map and Constituency Dashboard

The civic map (Fig. 4b) is an OpenStreetMap base layer rendered with Leaflet, as shown by the on-screen attribution, with filters for category, status and ward, a "My Location" control, a "Report Here" action and a status legend; the header reports 30 active pins across Chennai South. Selecting a pin opens the summary in Fig. 4c. The representative dashboard (Fig. 5) shows total complaints, resolution rate, average turnaround, citizen-reopened and escalated counts, a complaints-by-category bar chart, a status-distribution chart and a "Recurring Issue Hotspots" panel that reports 4 clusters detected by grouping complaints within 400 m. The clustering algorithm and the public (non-representative) view of the dashboard were not supplied.

G. Public Project and Fund Transparency

The specified project record contains name, location, responsible agency, category, approved amount and expenditure where available, dates, status (proposed, approved, in progress, completed, delayed or on hold) and a source reference, with a source state of officially verified, authority-submitted, citizen-reported or data unavailable. The system must not fabricate project or expenditure information; a college prototype uses clearly labelled sample data, and an issue is linked to a project only where available data support the relationship. The "follow the problem to the money" chain (issue, location, authority, project, allocation, expenditure, progress, verification, outcome) is a design concept. Only a "Public Projects" navigation entry was observed, so this module is marked U.

H. AI Assistant and Notifications

A "Civic Assistant" navigation entry and a notification bell with an unread badge were observed, but neither the assistant's behaviour nor the notification rules were supplied. The specification requires a rule-based fallback, optional use of a local open-source model, no paid AI dependency, and no AI authority over routing, status or official records. Notifications are to be in-app first, with browser or e-mail alerts optional. AI classification and duplicate detection are therefore treated as optional enhancements (Section VII).

I. Technology Stack and Implementation Status

Confirmed from the screenshots are a browser-based client, OpenStreetMap with Leaflet, bar and donut charts, and a Tamil-language toggle. The frontend framework, backend, database, authentication, storage and deployment are not confirmed. The free-only specification lists Next.js or React with Tailwind CSS, Next.js routes, FastAPI or Node.js, PostgreSQL (for example via Supabase or locally), Recharts or Chart.js and local or free-tier deployment as candidates; these are candidates, not implemented facts.

V. Experimental Evaluation and Results

A. Evaluation Objectives and Setup

Evaluation is separated into functional correctness, usability and real-world impact, because evidence for one does not establish the others. At the time of writing, the only evidence is a prototype walkthrough of the landing page, civic map, complaint summary and representative dashboard on 30 prototype complaint records for part of Chennai. The records are treated as sample data pending author confirmation and are not real-world findings. No test log, survey or participant record was supplied.

B. Functional Test Scenarios

The following scenarios are defined for the pilot build and have *not yet been executed or recorded*: S1 valid submission; S2 submission with a missing mandatory field; S3 unavailable location; S4 uncertain jurisdiction routed to review; S5 officer assignment; S6 status update recording role and timestamp; S7 "not resolved" leading to reopening; S8 escalation of an overdue or disputed case; S9 access attempt by an unauthorised role; and S10 public view exposing no phone number or private address.

C. Quantitative Measures

Metrics are limited to quantities that can be measured. With N_{pass} passed and N_{exec} executed test cases, $N_{correct}$ correctly routed cases among N_{known} cases with a known expected authority, and N_{done} participants completing a task among N_{try} attempting it:

$$PR = (N_{pass} / N_{exec}) \times 100 \quad (1)$$

$$RA = (N_{correct} / N_{known}) \times 100 \quad (2)$$

$$TCR = (N_{done} / N_{try}) \times 100 \quad (3)$$

Table IV gives the method, sample and observed result for each metric. Time to submit is the elapsed time from opening the report flow to successful submission; status comprehension is the share of participants who correctly explain a status or timeline view. Usability and trust measures require real participants, informed consent, a documented instrument and privacy-preserving data handling; none has been collected.

TABLE IV
EVALUATION METRICS, METHOD, SAMPLE AND OBSERVED RESULT

Metric	Method and sample	Observed result
Functional pass rate, Eq. (1)	Scenarios S1–S10 on the pilot build; test count [DATA REQUIRED]	Not measured
Routing accuracy, Eq. (2)	Reports with known expected authority; n [DATA REQUIRED]	Not measured
Reporting completion, Eq. (3); time to submit	Task sessions; participants [DATA REQUIRED]	Not measured
Status comprehension	Participants explain a status and timeline view; n [DATA REQUIRED]	Not measured
Verification-flow completion, Eq. (3)	Participants complete a verify or reopen task; n [DATA REQUIRED]	Not measured
Privacy and access control	Role-by-resource test matrix including public view	Not measured
Prototype data snapshot (demonstration data, not a performance result)	Values read from landing page and dashboard screenshots; 30 prototype records	30 records; 5 resolved; 1 reopened; 1 escalated; 4 hotspot clusters; 1.4 days average

D. Results and Interpretation

The prototype snapshot in Table IV is read from Fig. 4a and Fig. 5. Five of 30 records are shown as resolved, which is 16.7 % and is displayed as 17 %; one record was reopened by the

citizen, one is escalated, the average recorded turnaround is 1.4 days, and four recurring-issue clusters are reported. The category chart shows roads as by far the largest category among the seven displayed. These values demonstrate that the dashboard computes and displays aggregates from stored records; because the records are prototype data, they do not indicate real-world performance, and no inference about improvement in resolution or trust is made.

Two labelling inconsistencies were observed. The same 17 % figure is titled "Verified Resolution" on the landing page and "Resolution Rate" on the dashboard, and the 1.4-day figure is titled "Avg Response Time" on one screen and "Average Turnaround … time to resolve" on the other. The metric definitions should be unified and stated, and the verification-based figure should count only citizen-verified closures.

Resolved is a status, not proof that a problem was fixed, and a dashboard alone does not prove accountability.

E. Discussion

The evidence establishes that the map, complaint summary, aggregate dashboard and hotspot panel are demonstrable in the prototype and that the role-specific views differ. It does not establish routing correctness, usability, the behaviour of the officer, field-worker, verification and project modules, privacy enforcement, or any effect on citizens or departments. This matches the broader pattern in which few citizen-participation applications are evaluated by end users in real-world conditions [1]. The next step is to execute S1–S10, then a small task-based usability study with consent and de-identified records, before any pilot with a participating authority; success thresholds should be declared before testing rather than inferred afterwards.

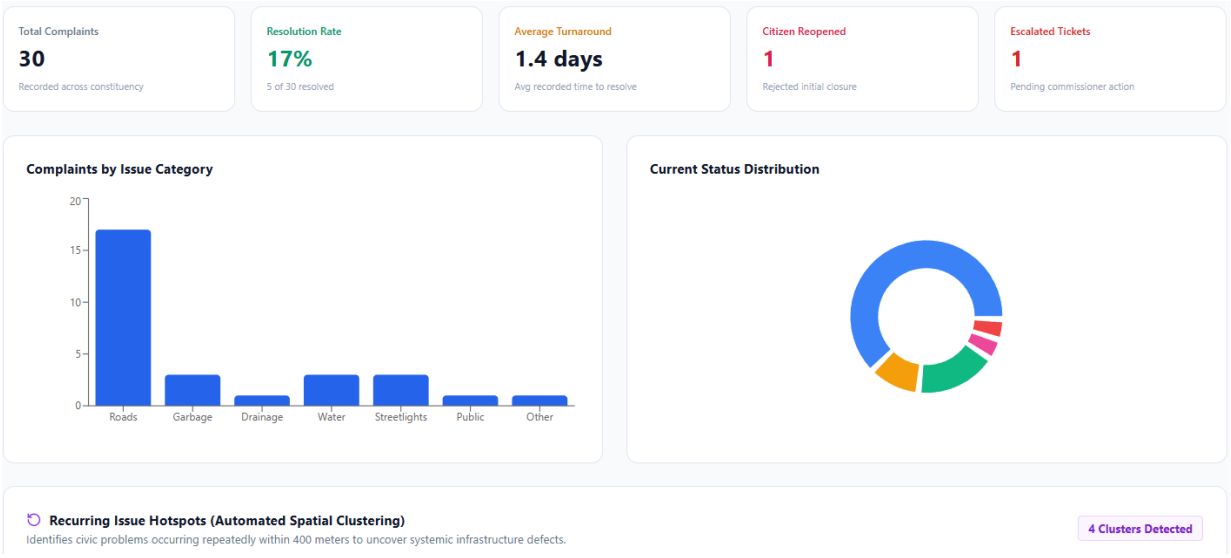


Fig. 5. Representative oversight dashboard (30 prototype complaint records): total complaints, resolution rate, average turnaround, citizen-reopened and escalated counts, complaints by category, status distribution and the recurring-issue hotspot panel (4 clusters detected, 400 m grouping). The page header containing the account name was cropped out.

VI. Advantages and Limitations

Potential advantages (design intentions, not proven results): the responsible authority is visible to the citizen; status history is traceable; a claimed resolution can be challenged and reopened; recurring problems are visible as clusters rather than isolated tickets; and project records, where verified, are understandable and source-labelled. The platform runs on free components and keeps AI optional.

Limitations: the system is a prototype without production deployment, evaluated only on a small prototype dataset in one area; routing depends on correct jurisdiction rules that must be configured from authoritative sources; official project and expenditure data may be incomplete or delayed, and no government API access is assumed; false, duplicate or abusive reports and location inaccuracies are possible; evidence retention raises privacy concerns; any AI component may err or be biased; officer, field-worker, verification and project modules are not evidenced here; and no long-term or representative user evaluation exists.

VII. Future Work

- Provide evidence for the officer, field-worker, verification, audit-log and project modules, and execute and record scenarios S1–S10.
- Validate routing rules against authoritative jurisdiction data and report routing accuracy on labelled cases.
- Run a consented, task-based usability study measuring completion, time to submit and status comprehension.
- Add duplicate detection and AI-assisted categorisation only with a labelled test set and human oversight.
- Import verified project and expenditure data, with provenance labels, using a structured open-data model such as that of [7].
- Explore asset-level repair history and recurring-failure analysis, before-and-after evidence with durability checks, and an explainable, evidence-based authority performance indicator used as an appraisal input; no popularity ranking or political scoring is intended. Volunteer assistance is outside the initial scope.

VIII. Conclusion

This paper presented CivicLens, a platform that connects civic issue reporting with visible authority routing, an auditable timeline, citizen verification with reopening and escalation, recurring-issue analytics and a source-labelled project-transparency design. A prototype demonstrates an OpenStreetMap-based civic map, a complaint summary and a representative dashboard with hotspot clustering on 30 prototype records. The evaluation to date is a walkthrough only: functional correctness, routing accuracy, usability and impact remain unmeasured, and a protocol is provided for measuring them. The value of the work will depend on completing and testing the remaining modules and on evidence from real users.

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